

- 1 Fig. 8.1 shows a sledge of mass 40 kg. It is being pulled across a horizontal surface of deep snow by a light horizontal rope. There is a constant resistance to its motion.

The tension in the rope is 120 N.



Fig. 8.1

The sledge is initially at rest. After 10 seconds its speed is 5 ms^{-1} .

- (i) Show that the resistance to motion is 100 N.

[4]

When the speed of the sledge is 5 ms^{-1} , the rope breaks.

The resistance to motion remains 100 N.

- (ii) Find the speed of the sledge

A 1.6 seconds after the rope breaks,

[3]

B 6 seconds after the rope breaks.

[1]

The sledge is then pushed to the bottom of a ski slope. This is a plane at an angle of 15° to the horizontal.

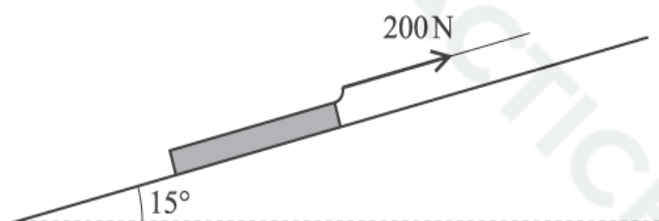


Fig. 8.2

The sledge is attached by a light rope to a winch at the top of the slope. The rope is parallel to the slope and has a constant tension of 200 N. Fig. 8.2 shows the situation when the sledge is part of the way up the slope.

The ski slope is smooth.

- (iii) Show that when the sledge has moved from being at rest at the bottom of the slope to the point when its speed is 8 ms^{-1} , it has travelled a distance of 13.0 m (to 3 significant figures).

[4]

When the speed of the sledge is 8 ms^{-1} , this rope also breaks.

- (iv) Find the time between the rope breaking and the sledge reaching the bottom of the slope.

[6]

- 2 Fig. 5 shows blocks of mass 4 kg and 6 kg on a smooth horizontal table. They are connected by a light, inextensible string. As shown, a horizontal force $F \text{ N}$ acts on the 4 kg block and a horizontal force of 30 N acts on the 6 kg block.

The magnitude of the acceleration of the system is 2 ms^{-2} .



Fig.5

- (i) Find the two possible values of F .

[4]

- (ii) Find the tension in the string in each case.

[3]

- 3 In this question, air resistance should be neglected.

Fig. 2 illustrates the flight of a golf ball. The golf ball is initially on the ground, which is horizontal.

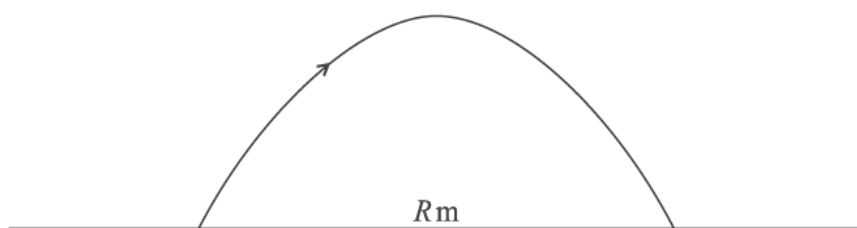


Fig.2

It is hit and given an initial velocity with components of 15 ms^{-1} in the horizontal direction and 20 ms^{-1} in the vertical direction.

- (i) Find its initial speed.

[1]

- (ii) Find the ball's flight time and range, $R \text{ m}$.

[4]

(iii)

- A Show that the range is the same if the components of the initial velocity of the ball are 20 ms^{-1} in the horizontal direction and 15 ms^{-1} in the vertical direction.

[1]

- B State, justifying your answer, whether the range is the same whenever the ball is hit with the same initial speed.

[2]

- 4 Fig. 7 illustrates a train with a locomotive, L, pulling two trucks, A and B.

The locomotive has mass 90 tonnes and is subject to a resistance force of 2000 N.

Each of the trucks A and B has mass 30 tonnes and is subject to a resistance force of 500 N.

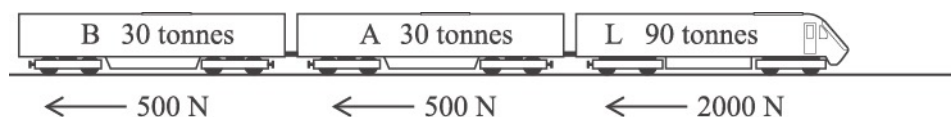


Fig. 7

Initially the train is travelling along a straight horizontal track. The locomotive is exerting a driving force of 12 000 N.

- (i) Find the acceleration of the train.

[3]

- (ii) Find the tension in the coupling between trucks A and B.

[3]

When the train is travelling at 10ms^{-1} , a fault occurs with truck A and the resistance to its motion changes from 500 N to 5000 N.

The driver reduces the driving force to zero and allows the train to slow down under the resistance forces and come to a stop.

- (iii) Find the distance the train travels while slowing down and coming to a stop.

Find also the force in the coupling between trucks A and B while the train is slowing down, and state whether it is a tension or a thrust.

[7]

The fault in truck A is repaired so that the resistance to its motion is again 500 N. The train continues and comes to a place where the track goes up a uniform slope at an angle of α° to the horizontal.

- (iv) When the train is on the slope, it travels at uniform speed. The driving force remains at 12 000 N. Find the value of α .

[3]

- (v) Show that the force in the coupling between trucks A and B has the same value that it had in part (ii).

[2]

- 5 In a science fiction story a new type of spaceship travels to the moon. The journey takes place along a straight line. The spaceship starts from rest on the earth and arrives at the moon's surface with zero speed. Its speed, v kilometres per hour at time t hours after it has started, is given by

$$v = 37500(4t - t^2).$$

- (i) Show that the spaceship takes 4 hours to reach the moon.

[1]

- (ii) Find an expression for the distance the spaceship has travelled at time t .

Hence find the distance to the moon.

[4]

- (iii) Find the spaceship's greatest speed during the journey.

[2]

- 6 A box of emergency supplies is dropped to victims of a natural disaster from a stationary helicopter at a height of 1000 metres. The initial velocity of the box is zero.

At time t s after being dropped, the acceleration, a m s⁻², of the box in the vertically downwards direction is modelled by

$$\begin{aligned} a &= 10 - t & \text{for } 0 \leq t \leq 10, \\ a &= 0 & \text{for } t > 10. \end{aligned}$$

- (i) Find an expression for the velocity, v m s⁻¹, of the box in the vertically downwards direction in terms of t for $0 \leq t \leq 10$.

Show that for $t > 10$, $v = 50$.



(ii) Draw a sketch graph of v against t for $0 \leq t \leq 20$.

[3]

(iii) Show that the height, h m, of the box above the ground at time t s is given, for $0 \leq t \leq 10$, by

$$h = 1000 - 5t^2 + \frac{1}{6}t^3.$$

Find the height of the box when $t = 10$.

[4]

(iv) Find the value of t when the box hits the ground.

[2]

(v) Some of the supplies in the box are damaged when the box hits the ground. So measures are considered to reduce the speed with which the box hits the ground the next time one is dropped. Two different proposals are made. Carry out suitable calculations and then comment on each of them.

A The box should be dropped from a height of 500 m instead of 1000 m.

[2]

B The box should be fitted with a parachute so that its acceleration is given by

$$\begin{aligned} a &= 10 - 2t & \text{for } 0 \leq t \leq 5, \\ a &= 0 & \text{for } t > 5. \end{aligned}$$

[3]

- 7 The battery on Carol and Martin's car is flat so the car will not start. They hope to be able to "bump start" the car by letting it run down a hill and engaging the engine when the car is going fast enough. Fig. 6.1 shows the road leading away from their house, which is at A. The road is straight, and at all times the car is steered directly along it.

- From A to B the road is horizontal.
- Between B and C, it goes up a hill with a uniform slope of 1.5° to the horizontal.
- Between C and D the road goes down a hill with a uniform slope of 3° to the horizontal. CD is 100 m. (This is the part of the road where they hope to get the car started.)
- From D to E the road is again horizontal.

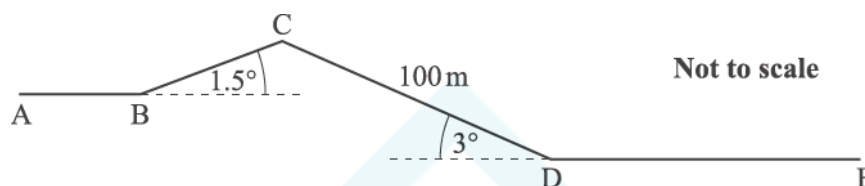


Fig. 6.1

The mass of the car is 750 kg, Carol's mass is 50 kg and Martin's mass is 80 kg.

Throughout the rest of this question, whenever Martin pushes the car, he exerts a force of 300 N along the line of the car.

- (i) Between A and B, Martin pushes the car and Carol sits inside to steer it. The car has an acceleration of 0.25 m s^{-2} .

Show that the resistance to the car's motion is 100 N.

[3]

Throughout the rest of this question you should assume that the resistance to motion is constant at 100 N.

- (ii) They stop at B and then Martin tries to push the car up the hill BC.

Show that Martin cannot push the car up the hill with Carol inside it but can if she gets out.

Find the acceleration of the car when Martin is pushing it and Carol is standing outside.

[6]

- (iii) While between B and C, Carol opens the window of the car and pushes it from outside while steering with one hand. Carol is able to exert a force of 150 N parallel to the surface of the road but at an angle of 30° to the line of the car. This is illustrated in Fig. 6.2.

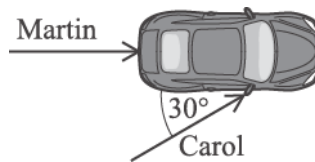


Fig. 6.2

Find the acceleration of the car.

[4]

- (iv) At C, both Martin and Carol get in the car and, starting from rest, let it run down the hill under gravity. If the car reaches a speed of 8 m s^{-1} they can get the engine to start.

Does the car reach this speed before it reaches D?

[5]

- 8 Fig. 4 illustrates a straight horizontal road. A and B are points on the road which are 215 metres apart and M is the mid-point of AB.

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When a car passes A its speed is 12 m s^{-1} in the direction AB. It then accelerates uniformly and when it reaches B its speed is 31 m s^{-1} .

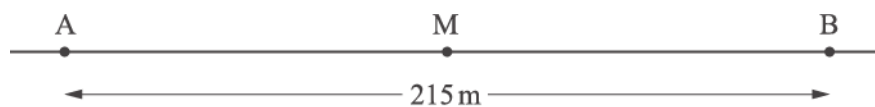


Fig. 4

- (i) Find the car's acceleration. [2]
- (ii) Find how long it takes the car to travel from A to B. [2]
- (iii) Find how long it takes the car to travel from A to M. [3]
- (iv) Explain briefly, in terms of the speed of the car, why the time taken to travel from A to M is more than half the time taken to travel from A to B. [1]

- 9 Fig. 2 shows a 6 kg block on a smooth horizontal table. It is connected to blocks of mass 2 kg and 9 kg by two light strings which pass over smooth pulleys at the edges of the table. The parts of the strings attached to the 6 kg block are horizontal.



Fig. 2

- (i) Draw three separate diagrams showing all the forces acting on each of the blocks.

[3]

- (ii) Calculate the acceleration of the system and the tension in each string.

[5]

10 Fig. 3.1 shows a block of mass 8 kg on a smooth horizontal table.

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This block is connected by a light string passing over a smooth pulley to a block of mass 4 kg which hangs freely. The part of the string between the 8 kg block and the pulley is parallel to the table.

The system has acceleration $a \text{ ms}^{-2}$.

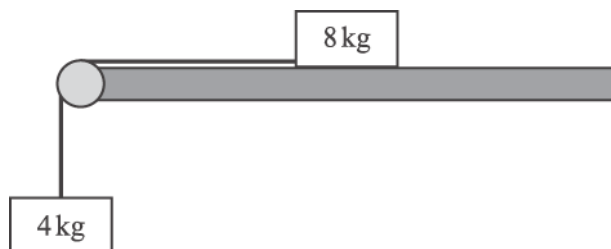


Fig. 3.1

(i) Write down two equations of motion, one for each block.

[2]

(ii) Find the value of a .

[2]

The table is now tilted at an angle of θ to the horizontal as shown in Fig. 3.2. The system is set up as before; the 4 kg block still hangs freely.

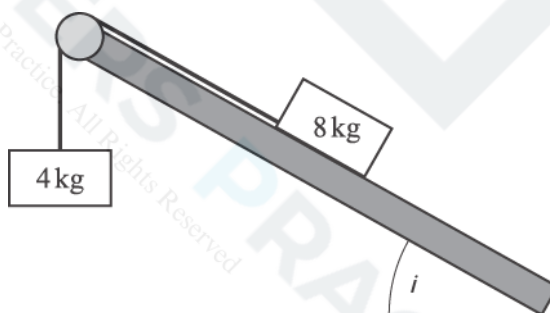


Fig. 3.2

(iii) The system is now in equilibrium. Find the value of θ .

[4]



Fig. 2

A particle moves on the straight line shown in Fig. 2. The positive direction is indicated on the diagram.

The time, t , is measured in seconds. The particle has constant acceleration, $a \text{ ms}^{-2}$.

Initially it is at the point O and has velocity $u \text{ ms}^{-1}$.

When $t = 2$, the particle is at A where OA is 12 m. The particle is also at A when $t = 6$.

(i) Write down two equations in u and a and solve them.

[4]

(ii) The particle changes direction when it is at B.

Find the distance AB.

[3]

- 12 A small package hangs from a balloon by means of a light inelastic string. The string is always vertical. The mass of the package is 15 kg.

Catherine initially models the situation by assuming that there is no air resistance to the motion of the package. Use Catherine's model to calculate the tension in the string if

- (a) the package is held at rest by the tension in the string, [1]
- (b) the package is instantaneously at rest and accelerating **upwards** at 2 m s^{-2} , [2]
- (c) the package is moving **downwards** at 3 m s^{-1} and accelerating **upwards** at 2 m s^{-2} . [1]

Catherine now carries out an experiment to find the magnitude of the air resistance on the package when it is moving. At a time when the package is accelerating **downwards** at 1.5 m s^{-2} , she finds that the tension in the string is 140 N.

- (d) Calculate the magnitude of the air resistance at that time. Give, with a reason, the direction of motion of the package. [5]

- 13 Fig. 3 shows a particle of weight 8 N on a rough horizontal table. The particle is being pulled by a horizontal force of 10 N. It remains at rest in equilibrium. EXAM PAPERS PRACTICE

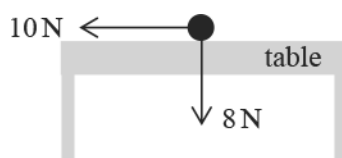


Fig. 3

- (a) What information given in the question tells you that the forces shown in Fig. 3 cannot be the only forces acting on the particle? [1]
- (b) The only other forces acting on the particle are due to the particle being on the table. State the types of these forces and their magnitudes. [2]

- 14 In an experiment a small box is hit across a floor. After it has been hit, the box slides without rotation. The box passes a point A. The distance the box travels after passing A before coming to rest is S metres and the time this takes is T seconds. The only resistance to the box's motion is friction due to the floor.
- The mass of the box is m kg and the frictional force is a constant F N.

(a) (i) Find the equation of motion for the box while it is sliding.

(ii) Show that $S = kT^2$ where $k = \frac{F}{2m}$.

[4]

(b) Given that $k = 1.4$, find the value of the coefficient of friction between the box and the floor.

[4]

- 15 Blocks A and B are connected by a light rigid horizontal bar and are sliding on a rough horizontal surface. A light horizontal string exerts a force of 40 N on B. This situation is shown in Fig. 14, which also shows the direction of motion, the mass of each of the blocks and the resistances to their motion.

side view

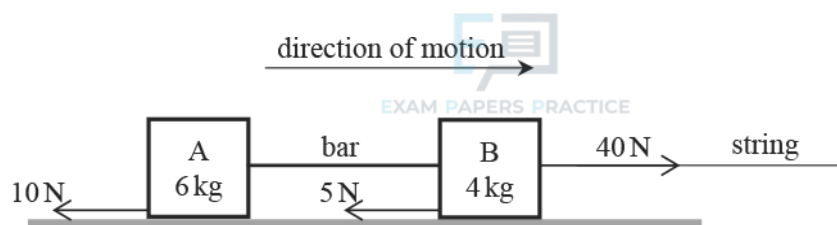


Fig.14

(a) Calculate the tension in the bar.

[4]

The string breaks while the blocks are sliding. The resistances to motion are unchanged.

(b) Determine

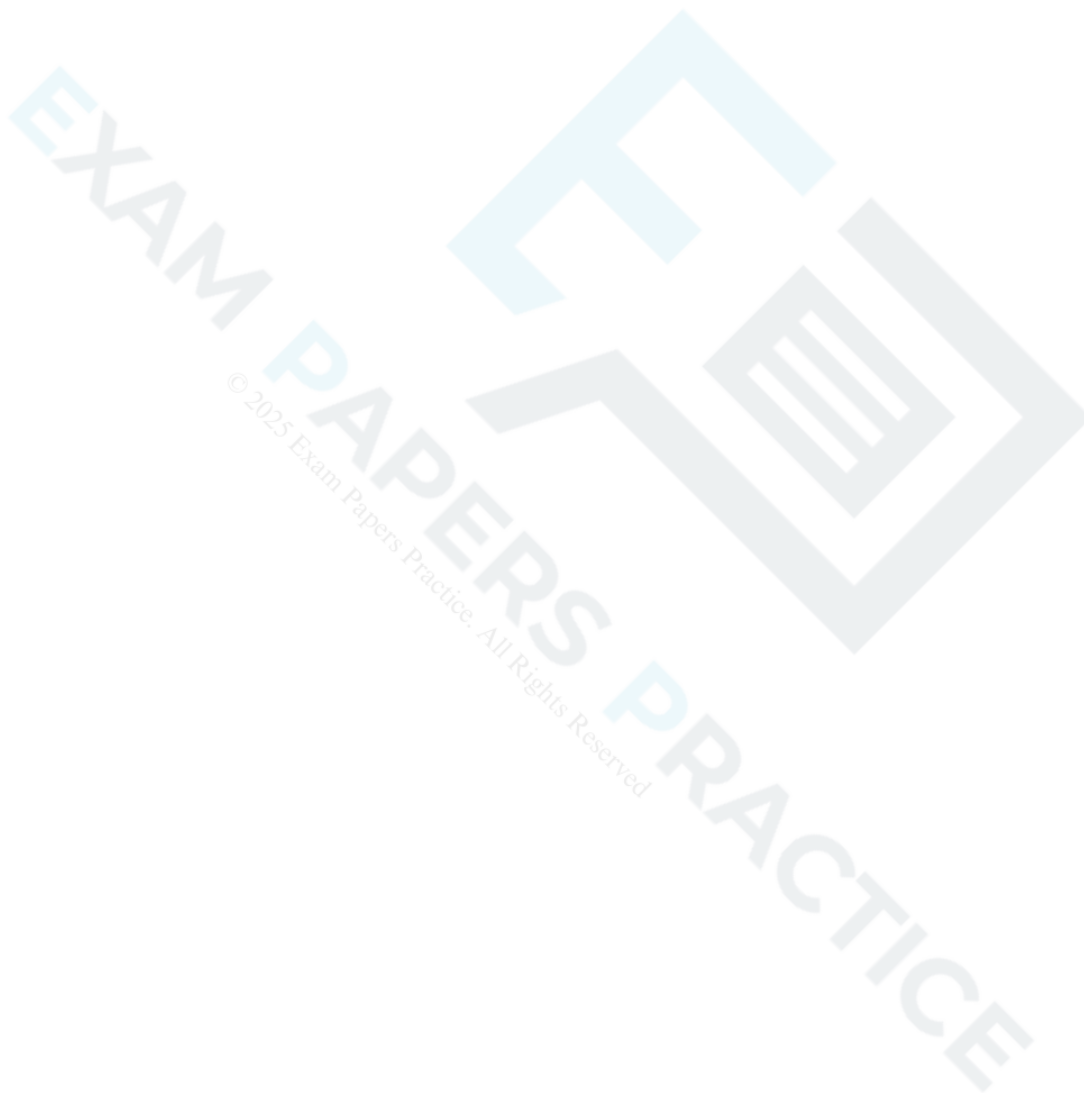
- the magnitude of the new force in the bar,
- whether the bar is in tension or in compression.

[5]

- 16 Dorothy and Harold push a wheelbarrow of mass 20 kg. They both push in the direction of motion. The wheelbarrow moves 10m from rest on a straight horizontal path. Dorothy applies a force of 90 N. Harold applies a force of 90 N for the first 3 m, and then reduces the force to 80 N for the remaining 7 m. The resistance to motion is 170 N throughout.

Find the time taken for the wheelbarrow to move the 10 m.

[9]



17 Austin pushes a curling stone of mass 18 kg on ice. He releases the stone and it slides 21 m in a straight line until coming to rest after 14 s. He models the motion by assuming that the frictional force is constant.

(a) Show that the speed of the stone at the moment that Austin releases it is 3 m s^{-1} .

[2]

(b) Find the magnitude of the frictional force acting on the stone.

[3]

(c) Calculate the coefficient of friction between the stone and the ice, giving your answer correct to 2 significant figures.

[3]

- 18 A car pulls a caravan along a straight road uphill on a slope that makes an angle of 8° with the horizontal. The mass of the car is 1700 kg and the mass of the caravan is 1300 kg. The driving force is 9000 N and the car and caravan experience resistances of 600 N and 1200 N respectively. The coupling between the car and the caravan is light and parallel to the road.
- (a) Draw a diagram showing all the forces acting on the car and all the forces acting on the caravan. [3]

- (b) Find the acceleration of the car and caravan as they move up the hill. [4]

- (c) Find the tension in the coupling between the car and the caravan. [3]

- 19 Fig. 4 shows two small blocks, Q of mass 8 kg and R of mass 6 kg. They are connected by a light string which passes over a pulley.

The pulley is light and smooth. It is rigidly suspended from the ceiling.

The system is released from rest with the two blocks at the same height.

Initially the blocks are 2 m above the floor and 3 m below the pulley.

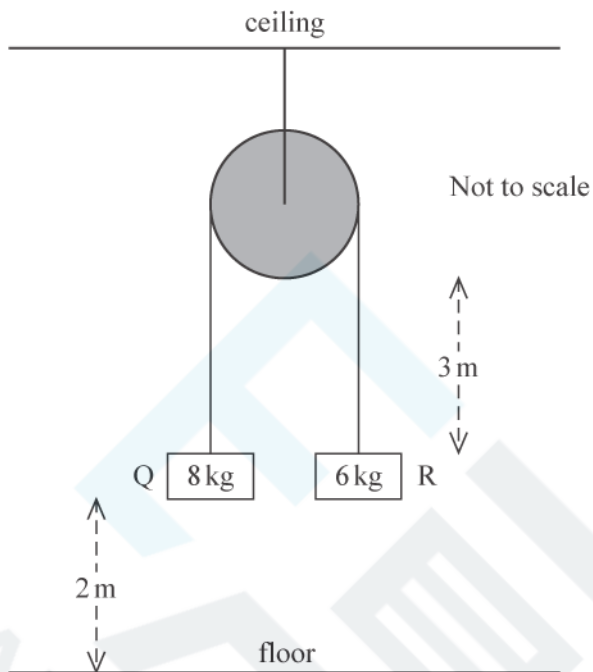


Fig. 4

- (i) Draw diagrams showing the forces acting on each of the blocks Q and R.

[1]

(ii) Write down the equations of motion of each of the blocks Q and R.

[2]

(iii) Find the time between the system being released and one of the blocks reaching the floor.

[4]

- 20 A block slides from rest down a line of greatest slope of a smooth plane inclined at 20° to the horizontal. Calculate the speed of the block when it has travelled 2 m.

[4]



- 21 Fig. 9 shows a block of mass 2 kg resting on a rough horizontal table. It is attached to a ball of mass $m\text{ kg}$ by a light inextensible string that passes over a smooth pulley at the edge of the table. The ball hangs vertically below the pulley. The coefficient of friction between the block and the table is 0.6 .

The system is held in equilibrium by a force of $5\sqrt{2}$ acting on the block at 45° above the horizontal. The block is on the point of sliding towards the pulley.

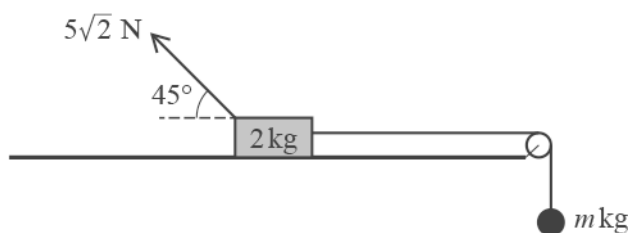


Fig. 9

- (a) Complete the force diagram above to show all the forces acting on the block and the ball.

[2]

(b) Calculate the frictional force acting on the block.

[4]

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(c) Calculate the value of m .

[4]

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- 22 Rory pushes a box of mass 2.8 kg across a rough horizontal floor against a resistance of 19 N . Rory applies a constant horizontal force. The box accelerates from rest to 1.2 ms^{-1} as it travels 1.8 m .

(a) Calculate the acceleration of the box.

[2]

(b) Find the magnitude of the force that Rory applies.

[2]

- 23 Fig. 4 shows a block of mass $4m$ kg and a particle of mass m kg connected by a light inextensible string passing over a smooth pulley. The block is on a horizontal table, and the particle hangs freely. The part of the string between the pulley and the block is horizontal. The block slides towards the pulley and the particle descends. In this motion, the friction force between the table and the block is $\frac{1}{2}mg$ N.

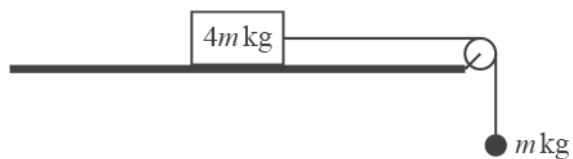


Fig. 4

Find expressions for

- the acceleration of the system,
- the tension in the string.

[4]

- 24 A model train consists of an engine of mass 0.3 kg and a truck of mass 0.2 kg. The train is pulled up a smooth plane inclined at 20° to the horizontal by a force of 2.5 N. This force is applied to the engine and acts parallel to a line of greatest slope of the plane.

(a) Show that the acceleration of the train is 1.65ms^{-2} correct to 3 significant figures. [3]

(b) Find the tension in the coupling between the engine and the truck. [2]

(c) Find the distance travelled by the train as it accelerates from 1ms^{-1} to 3ms^{-1} . [2]

- 25 A toy boat of mass 1.5 kg is pushed across a pond, starting from rest, for 2.5 seconds. During this time, the boat has an acceleration of 2 m s^{-2} . Subsequently, when the only horizontal force acting on the boat is a constant resistance to motion, the boat travels 10 m before coming to rest. Calculate the magnitude of the resistance to motion.

[6]



26(a) Fig. 6 shows a train consisting of an engine of mass 80 tonnes pulling two trucks each of mass 25 tonnes.

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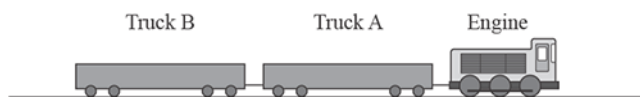


Fig. 6

Calculate the value of D .

[2]

- (b) The tension in the coupling between the engine and truck A is larger than that in the coupling between the trucks. Determine how much larger.

[2]

- 27 A car of mass 1200 kg travels from rest along a straight horizontal road. The driving force is 4000 N and the total of all resistances to motion is 800 N.

Calculate the velocity of the car after 9 seconds.

[4]

28(a) A block of mass 2 kg is placed on a rough horizontal table. A light inextensible string attached to the block passes over a smooth pulley attached to the edge of the table. The other end of the string is attached to a sphere of mass 0.8 kg which hangs freely.

The part of the string between the block and the pulley is horizontal. The coefficient of friction between the table and the block is 0.35. The system is released from rest.

Draw a force diagram showing all the forces on the block and the sphere.

[3]

(b) Write down the equations of motion for the block and the sphere.

[2]

(c) Show that the acceleration of the system is 0.35 m s^{-2} .

[4]

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(d) Calculate the time for the block to slide the first 0.5 m. Assume the block does not reach the pulley.

[2]

29 A projectile is fired from ground level at 35 m s^{-1} at an angle of θ° above the horizontal.

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Find the cartesian equation of the trajectory of the projectile.

[4]



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30 In this question, the x , and y directions are horizontal and vertically upwards respectively.

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A particle of mass 1.5 kg is in equilibrium under the action of its weight and forces $\mathbf{F}_1 = \begin{pmatrix} 4 \\ -2 \end{pmatrix} \text{ N}$ and $\mathbf{F}_2$.

The force $\mathbf{F}_2$ is changed to $\begin{pmatrix} 2 \\ 20 \end{pmatrix} \text{ N}$.

Find the acceleration of the particle.

[3]

END OF QUESTION PAPER

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Question			Answer/Indicative content	Marks	Guidance
1		i	$v = u + at$	M1	Use of a suitable constant acceleration formula
		i	$5 = 0 + a \times 10 \Rightarrow a = 0.5$	A1	Notice The value of a is not required by the question so may be implied by subsequent working
		i	$F = ma \Rightarrow 120 - R = 40 \times 0.5$	M1	Use of Newton's 2 nd Law with correct elements
		i	$R = 100\text{N}$	E1	Examiner's Comments ☐ ☐ This question involved a sledge being pulled, initially horizontally and then up a slope. Part (i) asked for the resistance to motion and required the use of a constant acceleration formula and then Newton's 2nd Law. It was very well answered. A few candidates lost marks by using the given final answer in an argument that was less than a valid verification.
		ii	(A) $F = ma \Rightarrow -100 = 40a$	M1	Equation to find a using Newton's 2 nd Law
		ii	$\Rightarrow a = -2.5$	A1	
		ii	When $t = 1.6$ $v = 5 + (-2.5) \times 1.6 = 1 \text{ ms}^{-1}$	A1	CAO Examiner's Comments This question involved a sledge being pulled, initially horizontally and then up a slope. The situation changed because the rope pulling the sledge broke. Candidates were asked to find the speed of the sledge at a time when it was still moving.

Question			Answer/Indicative content	Marks	Guidance
		ii	(B) When $t = 6$, it is stationary. $v = 0 \text{ ms}^{-1}$	B1	Examiner's Comments ☐ ☐ In part (ii) the situation changed because the rope pulling the sledge broke. In part (A) candidates were asked to find the speed of the sledge at a time when it was still moving and in part (B) at a later time when it would have come to a halt. Most candidates obtained the right answers to both parts. However, a few did not recognise that the acceleration changed when the rope broke and continued with the same value as they had in part (i). A more common mistake was to give a negative speed in part (B) rather than zero.
		iii	Motion parallel to the slope:	B1	Component of the weight down the slope, ie $40g \sin 15^\circ$ (= 101.457...)
		iii	$200 - 40g \sin 15^\circ = 40a$	M1	Equation of motion with the correct elements present. No extra forces.
		iii	$a = 2.463\dots$		This result is not asked for in the question
		iii	$v^2 - u^2 = 2as \Rightarrow 8^2 = 2 \times 2.46\dots \times s$	M1	Use of a suitable constant acceleration formula, or combination of formulae. Dependent on previous M1.
		iii	$\Rightarrow s = 12.989\dots$ rounding to 13.0 m	E1	Note If the rounding is not shown for s the acceleration must satisfy $2.452\dots < a < 2.471\dots$
		iv	Let a be acceleration up the slope		Examiner's Comments
		iv	$-40 \times 9.8 \times \sin 15^\circ = 40a$	M1	In part (iii) the sledge was being pulled up a smooth slope. There were many correct answers to this part but a few candidates were unable to use the component of the weight down the slope.
		iv	$a = -2.536\dots$, ie 2.536 ms^{-2} down the slope	A1	Use of Newton's 2 nd Law parallel to the slope
		iv	$s = ut + \frac{1}{2}at^2$		Condone sign error

Question			Answer/Indicative content	Marks	Guidance
		iv	$-12.989... = 8t + \frac{1}{2} \times (-2.536...)t^2$	M1	Dependent on previous M1. Use of a suitable constant acceleration formula (or combination of formulae) in a relevant manner.
		iv	$1.268...t^2 - 8t - 12.989... = 0$	A1	Signs must be correct
		iv	$t = \frac{8 \pm \sqrt{64 - 4 \times 1.268... \times (-12.989...)}}{2 \times 1.268...}$	M1	Attempt to solve a relevant three-term quadratic equation
		iv	$t = -1.339... \text{ or } 7.647... , \text{ so } 7.65 \text{ seconds}$	A1	
		iv	Alternative 2-stage motion		
		iv	Let a be acceleration up the slope		
		iv	$-40 \times 9.8 \times \sin 15^\circ = 40a$	M1	Use of Newton's 2nd Law parallel to the slope
		iv	$a = -2.536... , \text{ ie } 2.536 \text{ ms}^{-2} \text{ down the slope}$	A1	Condone sign error
		iv	Motion to highest point		
		iv	$v = u + at \Rightarrow 0 = 8 - 2.536...t$	M1	Dependent on previous M1. Use of a suitable constant acceleration formula, for either t or s , in a relevant manner.
		iv	$t = 3.154...$	A1	For either t or s
		iv	$s = ut + \frac{1}{2}at^2 \Rightarrow s = 8 \times 3.154... - \frac{1}{2} \times 2.536... \times 3.154...^2$		
		iv	$s = 12.616...$		
		iv	Distance to bottom = $12.989... + 12.616... = 25.605...$		
		iv	$s = ut + \frac{1}{2}at^2 \Rightarrow 25.605... = \frac{1}{2} \times 2.536... \times t^2$	M1	Use of a suitable constant acceleration formula
		iv	$t = 4.493...$		

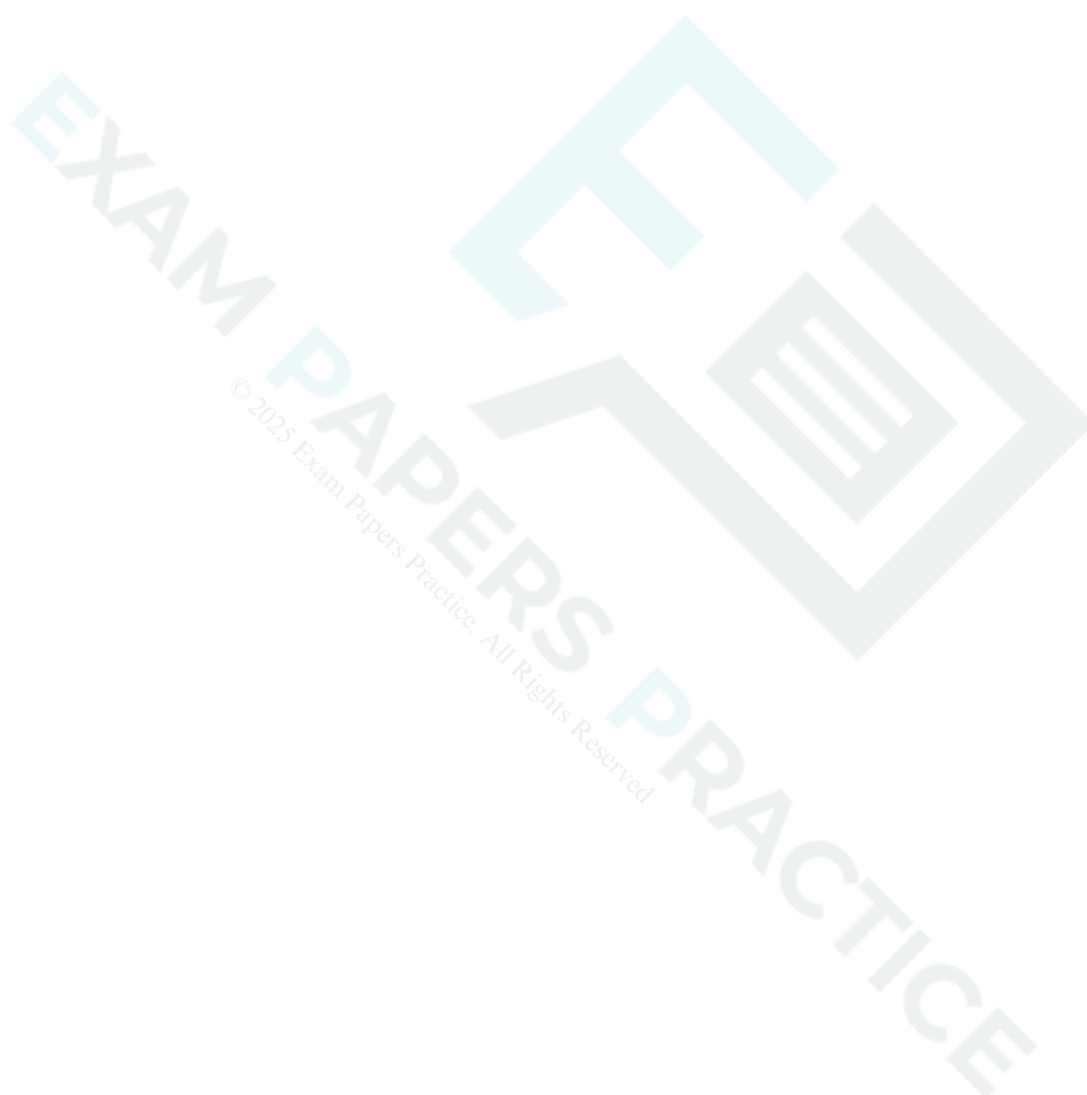
Question			Answer/Indicative content	Marks	Guidance
		iv	Total time = $3.154... + 4.493... = 7.647... \text{ s}$	A1	Examiner's Comments ☐ ☐ In part (iv), there was no longer a pulling force (the rope had broken again) and the sledge started moving up the slope, came to a stop and then slid down to the bottom of the slope. Candidates were asked to find how long this took. Many candidates knew what they had to do and there were plenty of correct answers; however, there were also many sign errors. This was the last question on the paper and several low-scoring candidates did not get started on it. There were also those who substituted completely wrong numbers into their constant acceleration formulae, indicating incorrect analysis of the situation.
			Total	18	

Question			Answer/Indicative content	Marks	Guidance
2		i	If the acceleration is to the right		
		i	Overall $30 - F = (4 + 6) \times 2$	M1	Newton's 2 nd Law in one direction. No extra forces allowed and signs must be correct.
		i	$F = 10$	A1	
		i	If the acceleration is to the left	M1	For considering second direction. No extra forces allowed and signs must be correct.
		i	$F = 50$	A1	<u>Examiner's Comments</u> This question was about connected particles, in the form of two blocks on a table. Part (i) was best answered treating the system as a whole. Both parts were correctly answered by many candidates. However, a few candidates did not realise that an acceleration of magnitude 2 could be in either direction, to the left or to the right.
		ii	6 kg block $30 - T = 6 \times 2$	M1	Newton's 2 nd law with correct elements on either block
		ii	$\Rightarrow T = 18$	A1	CAO No follow through from part (i)
		ii	In the other case $T = 42$	A1	CAO No follow through from part (i)
					<u>Examiner's Comments</u> This question was about connected particles, in the form of two blocks on a table. Part (ii) asked for the tension in the connecting string and so required candidates to work with one of the blocks. A not uncommon mistake, particularly in part (ii), was to introduce extra forces into the equations of motion.
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
3		i	Initial speed is 25 ms^{-1}	B1	Examiner's Comments This question was about a projectile (a golf ball). The horizontal and vertical components of its initial velocity were given. Nearly all candidates were able to find the initial speed and the flight time and range.
		ii	Vertical motion: $y = 20t - 4.9t^2$	M1	Forming an equation or expression for vertical motion
		ii	When $y = 0$,	M1	Finding t when the height is 0
		ii	$T = (0 \text{ or}) \frac{20}{4.9} = 4.08 \text{ s}$	A1	
		ii	$R = 15 \times 4.08... = 61.22$	F1	Allow $15 \times$ their T Note If horizontal and vertical components of the initial velocity are interchanged treat it as a misread; if no other errors are present this gives 3 marks.
		ii	Alternative Using time to maximum height		
		ii	Vertical motion: $v = 20 - 9.8t$	M1	Forming an equation or expression for vertical motion
		ii	Flight time = $2 \times$ Time to top	M1	Using flight time is twice time to maximum height or equivalent for range.
		ii	$T = 2 \times \frac{20}{9.8} = 4.08 \text{ s}$	A1	
		ii	$R = 15 \times 4.08... = 61.22$	F1	Allow $15 \times$ their T
		ii	Alternative Using formulae		
		ii	Finding angle of projection		
		ii	$\alpha = \arctan\left(\frac{20}{15}\right) = 53.1^\circ$	M1	Only award this mark if there is a clear intention to use this method
		ii	$R = \frac{2u^2 \sin \alpha \cos \alpha}{g} = \frac{2 \times 25^2 \times \sin 53.1^\circ \times \cos 53.1^\circ}{9.8}$	M1	Allow the alternative form $R = \frac{u^2 \sin 2\alpha}{g}$ with substitution
		ii	$R = 61.2$	A1	

Question			Answer/Indicative content	Marks	Guidance
		ii	$T = \frac{2u \sin \alpha}{g} = 4.08$	A1	Examiner's Comments Common mistakes were to interchange the vertical and horizontal components, and, for those who used the method of finding the time to maximum height, to fail to double it for the flight time.
		iii	(A) Flight time = $\frac{15}{4.9}$		
		iii	Range = $20 \times \frac{15}{4.9} = 61.22$	B1	Allow FT from part (ii) for a correct argument that they should be the same Examiner's Comments In part (iii) (A), candidates were asked to show that the range was the same if the components of the initial speed were interchanged; most did this by repeating the calculation from part (ii) but a few saw that this result could be deduced from the form of the expression for the range.
		iii	(B) No	M1	Attempt at disproof or counter-example. There must be some reference to the angle.
		iii	eg angle of projection 45°	A1	Complete argument Examiner's Comments Candidates went into part (iii) (B) having just met an example where the same initial speed but a different angle of projection produced the same range; they were asked whether this was generally true. Many candidates saw the point of the question and gave a counter-example (commonly the ball being projected vertically upwards). However, others incorrectly thought that the statement was generally true. There were also many answers which gave an inadequate explanation of the correct result.

Question			Answer/Indicative content	Marks	Guidance
			Total	8	



Question			Answer/Indicative content	Marks	Guidance
4		i	Whole train: mass = 150 tonnes	B1	Both totals required. Correct elements must be present CAO. Errors with units (eg not converting tonnes to kilograms) are penalised here but condoned where possible for the remainder of the question. Examiner's Comments This question was a good source of marks for many candidates. It was about the motion of a train and the forces in one of the couplings. Most candidates were able to answer the parts that involved considering the train as a whole. Fewer were successful when it came to working with part of the train to find the force in a particular coupling. Candidates were asked find the acceleration of the whole train and most were successful. A number failed to convert the mass of the train from tonnes into kilograms. This was penalised here but follow-through was then applied for all the marks in the next two parts and for the first two marks in part (iv).
		i	Total Resistance = 3000 N		
		i	$12000 - 3000 = 150000a$	M1	
		i	$a = 0.06$ The acceleration is 0.06 ms^{-2}	A1	
		ii	Truck B: $T - 500 = 30000a$	M1	Correct elements must be present Allow FT for a from part (i) if units are used consistently, for all the marks in this part
		ii	$T - 500 = 30000 \times 0.06$	A1	
		ii	$T = 2300$	A1	
		ii	Between A and B, tension of 2300 N		
		ii	Alternative		

Question			Answer/Indicative content	Marks	Guidance
		ii	Rest of train: $12\,000 - 2500 - T = 120\,000a$	M1	Correct elements must be present Examiner's Comments This question was a good source of marks for many candidates. It was about the motion of a train and the forces in one of the couplings. Most candidates were able to answer the parts that involved considering the train as a whole. Fewer were successful when it came to working with part of the train to find the force in a particular coupling. Candidates were asked to find the tension in the coupling between the two trucks. Most candidates answered this correctly but some introduced extraneous forces.
		ii	$T = 12\,000 - 2500 - 120\,000 \times 0.06$	A1	
		ii	$T = 2300$	A1	
		iii	Treating the train as a whole $-2000 - 5000 - 500 = 150\,000a$	M1	Allow FT for the remaining A marks in part (iii) from an error in a Correct elements must be present. Alternative for rest of train: $-T - 5000 - 2000 = 120\,000 \times -0.05$ The sign of 1000 must be consistent with the direction of T. Dependent on previous M and A marks. Accept "compression".
		iii	$a = -0.05$	A1	
		iii	$v^2 - u^2 = 2as$	M1	
		iii	$0^2 - 10^2 = 2 \times (-0.05) \times s$		
		iii	$s = 1000$ Stopping distance is 1000 m	A1	

Question			Answer/Indicative content	Marks	Guidance
		iii	B: $T - 500 = 30000a$	M1	Correct elements must be present. Alternative for rest of train: $-T - 5000 - 2000 = 120\,000 \times -0.05$
		iii	$T = -1000$	A1	The sign of 1000 must be consistent with the direction of T .
		iii	Between A and B, thrust of 1000 N	A1	Dependent on previous M and A marks. Accept "compression".
Examiner's Comments This question was a good source of marks for many candidates. It was about the motion of a train and the forces in one of the couplings. Most candidates were able to answer the parts that involved considering the train as a whole. Fewer were successful when it came to working with part of the train to find the force in a particular coupling. Carried 7 marks. The first four of these involved the motion of the train as a whole in a new situation and many candidates obtained all of these marks. The last three marks were for finding the new force in the coupling between the trucks. Most of those candidates who had been successful in part (ii) obtained these marks but others were unable to identify which forces were relevant and which were not.					
		iv	Equilibrium parallel to the slope	M1	Correct elements must be present and there must be an attempt to resolve the weight. Condone omission of g for this mark.
		iv	$150000 \times 9.8 \times \sin \alpha + 3000 = 12000$	A1	

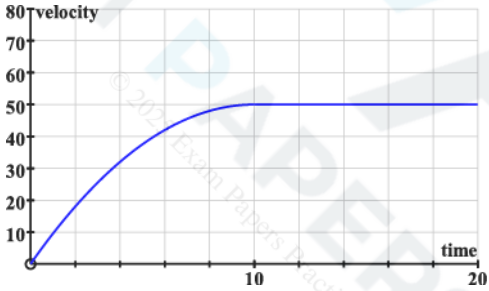
Question			Answer/Indicative content	Marks	Guidance
		iv	$\alpha = 0.35^\circ$	A1	CAO Examiner's Comments This question was a good source of marks for many candidates. It was about the motion of a train and the forces in one of the couplings. Most candidates were able to answer the parts that involved considering the train as a whole. Fewer were successful when it came to working with part of the train to find the force in a particular coupling. Involved a new situation in which the train was on a slope. Candidates were asked to find the angle of the slope. While there were plenty of correct answers, many of them were not very well explained. Good force diagrams were something of a rarity.
		v	B: $T_2 - 500 - 30000 \times 9.8 \times \sin 0.35\dots^\circ = 0$	M1	Correct elements must be present. Condone omission of g for this mark. Do not accept 1800 N for the component of the weight without justification. Alternative for rest of train: $12\,000 = T + 2500 + 120\,000 \times 9.8 \times \sin 0.35^\circ$

Question			Answer/Indicative content	Marks	Guidance
		v	$T_2 = 2300$ Between A and B, tension of 2300 N, as in part (ii)	A1	This mark can only be awarded if the angle found in (iv) is correct. Examiner's Comments This question was a good source of marks for many candidates. It was about the motion of a train and the forces in one of the couplings. Most candidates were able to answer the parts that involved considering the train as a whole. Fewer were successful when it came to working with part of the train to find the force in a particular coupling. Provided the last two marks on the paper. It exemplified the interesting (and little known) result that the tensions between trucks when the train is going up a slope at constant speed are the same as those when it is accelerating on level ground, under the same driving force. In this part candidates were asked to do no more than find the same numerical answer as they had obtained in part (ii). Stronger candidates were successful on this but many others had failed to find the correct earlier results (the tension in part (ii) and the angle in part (iv)) that were needed here.
			Total	18	

Question			Answer/Indicative content	Marks	Guidance
5		i	$v = 0$ when it arrives	B1	Award this mark for substituting $t = 4$ to obtain $v = 0$
		i	$150\,000(t - \frac{1}{4}t^2) = 0$	B1	Award this mark for substituting $t = 4$ to obtain $v = 0$
		i	$\Rightarrow t = 4$ (on arrival)		Condone omission of $t = 0$
		ii	Distance travelled $s = \int v dt$	M1	Do not accept multiplication by t .
		ii	$s = 150\,000 \left[\frac{1}{2}t^2 - \frac{1}{12}t^3 \right] (+c)$	A1	
		ii	When $t = 4$, $s = 400\,000$	M1	Substituting their $t = 4$. This mark is dependent on the previous M mark.
		ii	The journey is 400 000 km	A1	If 400 000 seen award the previous mark
		iii	For maximum speed $a = \frac{dv}{dt} = 0$		
		iii	$\frac{dv}{dt} = 150\,000(1 - \frac{1}{2}t)$		
		iii	$\rightarrow t = 2$ seen	B1	$t = 2$ seen Accept a trial and error method
		iii	$v = 150\,000(2 - \frac{1}{4} \times 2^2) = 150\,000$		
		iii	Maximum speed is 150 000 km h ⁻¹	B1	CAO
					Examiner's Comments This question involved motion with non-uniform acceleration. It was set in the context of a science fiction flight to the moon. It was well answered with many candidates obtaining full marks. Only a few candidates made the mistake of attempting to use constant acceleration formulae. There were some mistakes in the integration to go from speed to distance; an extraneous t appeared in some scripts.

Question			Answer/Indicative content	Marks	Guidance
			Total	7	



Question			Answer/Indicative content	Marks	Guidance
6		i	Integrate a to obtain v	M1	Attempt to integrate Substitution of $t = 10$ to find v Sound argument required for given answer. It must in some way refer to $a = 0$. Examiner's Comments Candidates were asked to find the velocity in each stage of the motion. This required the use of calculus and was quite well-answered. A common mistake was to omit the explanation that the terminal velocity is constant because the acceleration is known to be zero.
		i	$v = 10t - \frac{1}{2}t^2 \quad (+c)$	A1	
		i	$t = 10 \Rightarrow v = 100 - 50 = 50$	M1	
		i	Since $a = 0$ for $t > 10$, $v = 50$ for $t > 10$	A1	
		ii	Continuous two part $v-t$ graph 	B1	The graph must cover $t = 0$ to $t = 20$ B1 B1 B0 if no vertical scale is given Examiner's Comments Involved drawing a velocity-time graph for the two stages of the motion. This was well-answered but some candidates gave a straight line rather than a curve for the first stage.
		ii	Curve for $0 \leq t \leq 10$	B1	
		ii	Horizontal straight line for $10 \leq t \leq 20$	B1	
		iii	Distance fallen = $\int \left(10t - \frac{1}{2}t^2 \right) dt$	M1	Attempt to integrate A1
		iii	$d = 5t^2 - \frac{1}{6}t^3 + c \quad (c = 0)$	A1	
		iii	Height = $1000 - d$		

Question			Answer/Indicative content	Marks	Guidance
		iii	Height = $1000 - 5t^2 + \frac{1}{6}t^3$	A1	This mark should only be given if the signs are correctly obtained. oe Examiner's Comments This part involved going from the velocity of the box to its height and so involved further use of calculus. That part was well-done but many candidates failed to deal adequately with the sign change involved in going from the distance fallen from the helicopter to the height above the ground. <u>Marking guidance</u> Follow through between parts of this question should be allowed for the value of h (when $t = 10$) found in part (iii) if it is used in part (iv) or in part (v)(A).
		iii	When $t = 10$, $h = 667$	B1	
		iv	Time at constant vel = $667 \div 50 = 13.3$	B1	FT for h from part (iii) FT Examiner's Comments Candidates were asked for the time taken for the box to fall. Only a few candidates got this right. Almost all forgot that there were two stages to the motion. Many set the cubic expression for height equal to zero and then tried to solve the resulting cubic equation. The correct approach was nothing like as difficult as that.
		iv	Total time $t = 10 + 13.3 = 23.3$	B1	
		v	(A) Since $500 > 333$	M1	For finding the height at which the crate reaches terminal velocity, eg $h = 167$, or equivalent relevant calculation. FT for h from part (iii) if used. Allow either one (or both) of these two statements. Integration to find v
		v	The box will have reached terminal speed. So there is no improvement	A1	
		v	(B) $v = 10t - t^2$ (for $t \leq 5$)	M1	
		v	Terminal velocity is 25 ms^{-1}	A1	

Question			Answer/Indicative content	Marks	Guidance
		v	So better	A1	Examiner's Comments Two alternative methods of trying to limit the damage to the box on landing were considered. The first, in part (A), involved reducing the height from which the box was dropped; however, it would still have reached terminal velocity so this would not have helped. This was not answered very well with many candidates missing the point. The second, in part (B), involved fitting a parachute and this was very well answered, with many candidates obtaining full marks on it. Overall this question was not answered badly. It was not difficult but a high mark on it did require clear thinking.
			Total	18	

Question			Answer/Indicative content	Marks	Guidance
7		i	$F - R = ma$	M1	Use of Newton's 2 nd Law
		i	$300 - R = (750 + 50) \times 0.25$	A1	Correct elements present
		i	$R = 100$	A1	This is a given result
					Examiner's Comments Generally this question was answered very well and many candidates got it completely right. Almost all candidates were successful on this part involving the use of Newton's 2 nd Law. Marking guidance Follow through between parts of this question should be allowed for values found in parts (ii) and (iii) providing the questions are not simplified.
		ii	Carol in Component of weight down slope	M1	Resolving down the slope. Accept use of 750 instead of 800.
					For this mark only condone no g and allow sin-cos interchange.
		ii	$= 800g \sin 1.5^\circ (= 205.2 \text{ N})$	A1	Give M1 A1 for $800g \sin 15^\circ$ seen
		ii	Martin has to overcome 305.2 N		
		ii	$300 < 305.2$ Martin cannot manage	A1	This mark may be awarded for an argument based on Newton's 2 nd law leading towards $a = -0.006$
		ii	Carol out Martin has to overcome $750g \sin 1.5^\circ + 100 = 292.4 \text{ N}$		
		ii	$300 > 292.4$ so Martin manages	B1	Explanation, based on correct working, that Martin can manage. This can be given retrospectively with a comment on a positive value for a .
		ii	$300 - 292.4 = 7.6 = 750a$	M1	Use of Newton's 2 nd Law

Question			Answer/Indicative content	Marks	Guidance
		ii	The acceleration is 0.010 ms^{-2}	A1	Cao. Accept 0.01 or an answer that rounds to 0.01. Examiner's Comments Was about pushing a car up a slope. This was well-answered with most candidates resolving the weight correctly to find its component down the slope. Very few made the mistake of interchanging sin and cos.
		iii	Component of Carol's force parallel to the line of the car	M1	For attempt at resolution in the correct direction. For this mark only, condone sin-cos interchange.
		iii	$= 150\cos 30^\circ (= 129.9)$	A1	Give M1 A1 for $150\cos 30^\circ$ seen
		iii	Resultant forward force = $7.6 + 129.9 = 137.5$ $750a = 137.5$	M1	All forces parallel to the slope present and correct. Sign errors condoned.
		iii	The acceleration is 0.183 ms^{-2}	A1	FT their force parallel to the slope from part (ii) (correct value 7.6 N) Examiner's Comments The car was still being pushed up the slope but an extra force was now involved. This meant that the equation of motion had five terms and some candidates lost marks by omitting one (or more) of them.
		iv	Component of weight down the slope $= (750 + 50 + 80) \times 9.8 \times \sin 3^\circ$		
		iv	$880a = 451.3 - 100$	M1	Newton's 2nd law with correct elements present. No sin-cos interchange. The same mass must be used in both places.
		iv	$a = 0.399$	A1	
			$v^2 - u^2 = 2as$		
		iv	When $v = 8$, $s = 8^2 \div (2 \times 0.399)$	M1	Selection and use of an appropriate formula (unless with $a = g$)
		iv	$s = 80.1$	A1	FT their value of a

Question			Answer/Indicative content	Marks	Guidance
		iv	80.1 < 100 so Yes they get the car started	A1	FT their value of a
		iv	Alternative: Finding the speed after 100 m		
			Component of weight down the slope		
		iv	$= (750 + 50 + 80) \times 9.8 \times \sin 3^\circ$	M1	Newton's 2nd law with correct elements present. No sin-cos interchange
		iv	$880a = 451.3 - 100$		
		iv	$a = 0.399$	A1	
			$v^2 - u^2 = 2as$		
		iv	$v^2 = (0^2) + 2 \times 0.399 \times 100$	M1	Selection and use of an appropriate formula (unless with $a = g$)
		iv	$v = (\sqrt{79.8}) = 8.93...$	A1	FT their value of a
		iv	$(v > 8)$ so they get the car started	A1	FT their value of a
					Examiner's Comments
					The car travelled down a slope and many candidates were able to select the right information and methods to deal with this new situation. Some lost marks by forgetting one of the terms, for example the resistance.
					Overall the response to this question was very pleasing.
			Total	18	

Question			Answer/Indicative content	Marks	Guidance
8		i	$v^2 - u^2 = 2as$		
		i	$31^2 - 12^2 = 2 \times 215 \times a$	M1	Selection and use of appropriate equation(s)
		i	$a = 1.9$ so 1.9 ms^{-2}	A1	
		ii	$v = u + at$		
		ii	$31 = 12 + 1.9t$	M1	Selection and use of appropriate equation(s)
		ii	$t = 10$ so 10s	A1	FT from their value of a from part (i).
		iii	$s = ut + \frac{1}{2}at^2$		
		iii	$\frac{215}{2} = 12t + \frac{1}{2} \times 1.9 \times t^2$	M1	Selection and use of $s = ut + \frac{1}{2}at^2$, oe.
		iii	$\left(t = \frac{-12 \pm \sqrt{12^2 + 4 \times 0.95 \times 107.5}}{1.9} \right)$	M1	Correct elements but condone minor arithmetic errors.
		iii	$t = 6.055$ (or -18.69)	A1	Use of quadratic formula (may be implied by answer), oe.
		iii	Alternative: Finding a 2-stage method		FT their a only.
		iii	$v^2 - u^2 = 2as$ and $s = \frac{(u+v)}{2}t$		
		iii	$v = \pm \sqrt{12^2 + 2 \times 1.9 \times 107.5} = (\pm)23.505...$	M1	Selection and use of a complete valid 2-stage method
		iii	$s = \frac{(u+v)}{2}t \Rightarrow t = \frac{2 \times 107.5}{(12 + 23.505...)} \left(\text{or } t = \frac{2 \times 107.5}{(12 - 23.505...)} \right)$	M1	Using the output from the first stage to find t
		iii	$t = 6.055$ (or 18.69)	A1	FT their a only.

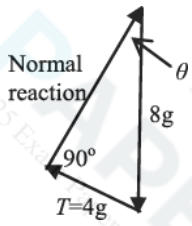
Question			Answer/Indicative content	Marks	Guidance
		iv	Because it is accelerating, it travels less fast in the first half of the distance and so takes more time.	B1	The answer must refer to the two parts of the distance (or “the same distance”) so no credit is given to answers like “Because it is accelerating” and “Because its speed is not uniform”. Most successful answers will refer to the times to cover AM and MB but this may be implicit. So B1 should be given for an answer like “It is travelling faster between M and B than it is between A and M” Notice that the fact that the acceleration is uniform is irrelevant. Examiner's Comments This question, about the motion of a car on a road, was very well answered and nearly all candidates got the first 7 marks for selecting and using appropriate constant acceleration formulae. The final part, for 1 mark, asked them to explain why the time to the mid-point of the distance is more than half the time for the whole journey (the car was accelerating throughout). This produced a number of excellent answers but many were unconvincing. A good answer made the point that the distance is the same in both parts but, because it is accelerating, the car is travelling slower in the first part than in the second part, and so takes more time.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
9		i		B1	Diagrams for both 2 and 9 kg blocks. The tensions must be different from each other. No extra forces.
		i		B1	Tensions on 6 kg block. The tensions must be different from each other. No extra forces.
		i		B1	6g and R on 6 kg block. No extra forces.
					Special Case When the tensions are given as T_1, T_2, T_3, T_4 (or equivalent) award up to SC1 SC0 for the first two marks. Examiner's Comments Candidates were asked to draw separate diagrams showing the forces on the three objects. There were two strings and the tensions in them were different. However, many candidates treated them as the same and just marked them as T. This was obviously a serious mistake which undermined the whole question and so caused a large loss of marks both in part (i) and, if continued.
		ii	$9g - T_2 = 9a$	M1	First equation correct
		ii	$T_2 - T_1 = 6a$	M1	Both the remaining two equations correct.
		ii	$T_1 - 2g = 2a$		Do not give this mark if both tensions are shown as the same.
		ii	$a = \frac{7}{17}g = 4.04 \text{ (m s}^{-2}\text{)}$	A1	The final three marks are dependent on both M marks a , T_1 and T_2 may be found in any order and FT should be allowed from the first of these found
		ii	$T_1 = 27.7 \text{ (N)}$	A1	
		ii	$T_2 = 51.9 \text{ (N)}$	A1	
		ii	Alternative: Whole system		
		ii	$9g - 2g = 17a$	M1	

Question			Answer/Indicative content	Marks	Guidance
		ii	$a = \frac{7g}{17} = 4.04$	A1	Both equations correct. Oe. The final two marks are dependent on both M marks. T_1 and T_2 may be found in either order and FT should be allowed from their value for a. Examiner's Comments Candidates went on to find the acceleration of the system and the tensions in the two strings. Those who had drawn correct diagrams in part (i) usually got this completely correct but those who had not done so, rarely made much progress on this part.
		ii	$T_1 - 2g = 2a$ and $9g - T_2 = 9a$	M1	
		ii	$T_1 = 27.7 \text{ (N)}$	A1	
		ii	$T_2 = 51.9 \text{ (N)}$	A1	
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
10		i	$T = 8a$	B1	Allow if a is in the upwards direction but the two equations must be consistent in this. Examiner's Comments Question 3 involved two connected particles in two different situations. In part (i) candidates were asked to write down the equation of motion of each particle. Most did this correctly but a common mistake was to introduce the weight of the block that was on a smooth horizontal table as an extra force.
		i	$4g - T = 4a$	B1	
		ii	Adding the two equations $\square 4g = 12a$	M1	Or equivalent method. No FT from part (i). CAO but allow 3.26. Examiner's Comments The question then went on to ask candidates to solve the equations to find the acceleration of the system. Those who got the right equations in the previous part were almost entirely successful. By contrast those who made a mistake on one or both equations in part (i) were almost entirely unsuccessful. No follow through was allowed from wrong equations in part (i). Answer 3.27 ms^{-2}
		ii	$a = \frac{g}{3} (-3.27 \text{ ms}^{-2})$	A1	
		iii	Equilibrium equations		Vertical equation Award if $8g \sin \theta$ seen. Do not allow sin-cos interchange Correct equation with $T = 4g$ substituted Note Award M1 M1 A1 for going straight to $4g = 8g \sin \theta$ oe Allow M1 M1 A0 for $4 = 8 \sin \theta$ with no previous work
		iii	$T - 4g = 0$	M1	
		iii	$T - 8g \sin \theta = 0$	M1	
		iii	$4g - 8g \sin \theta = 0$	A1	
		iii			



Question			Answer/Indicative content	Marks	Guidance
		iii	$\square \theta = 30^\circ$	A1	CAO
		iii	Alternative	M1	Examiner's Comments In part (iii) the table was titled and the system was in equilibrium. Candidates were asked to find the angle of the table. There were many correct answers. The most common mistake was to try to work with the weight of the block on the table rather than its resolved component down the slope. A few candidates lost a mark by missing g out altogether. Answer 30°
		iii	$T - 4g = 0$	M1	
		iii	Triangle of forces for the 8 kg block	M1	
		iii		A1	
		iii	$\sin \theta = \frac{4g}{8g}$	A1	Dependent on both M marks.
		iii	$\theta = 30^\circ$	A1	CAO
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
11		i	$s = ut + \frac{1}{2} at^2$		The final mark scheme will include commonly used alternative methods.
		i	$t = 2 \quad 2u + 2a = 12$	B1	Allow one equation with $v = 0$ and $t = 4$
		i	$t = 6 \quad 6u + 18a = 12$	B1	
		i	Solving the simultaneous equations	M1	Attempt to solve non-trivial simultaneous equations in u and a
		i	$u = 8, a = -2$	A1	CAO
					Examiner's Comments This question was about the movement of a particle along a straight line with constant acceleration. In part (i) candidates were asked to use the given information to find two equations for the initial velocity and the acceleration and to solve them. Most candidates got this right. Only a few failed to find the equations and there were also some careless mistakes when it came to solving them. Answer $u = 8, a = -2$
		ii	At B, $v^2 - u^2 = 2as$		Follow through for their values of u and a .
		ii	$0^2 - 8^2 = 2 \times -2 \times s$	M1	Allow the use of $s = ut + \frac{1}{2} at^2$ with $t = 4$.
		ii	$s = 16$	A1	
		ii	AB is 4 m.	A1	CAO
					Examiner's Comments The question then went on to ask for a distance AB where B was the point where the particle was instantaneously at rest. Most candidates successfully found the distance OB but a common mistake was to fail to subtract OA to find the distance requested. Answer 4 m
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
12		a	No acceleration so we require the weight 15g (147 N)	B1(AO3.3) [1]	Accept 15g, 15g N, 147 N etc
		b	Tension T N, $N2L \uparrow T - 147 = 15 \times 2$ So $T = 177$ and tension is 177 N	M1(AO3.4) A1(AO1.1) [2]	Application of N2L
		c	177 N	B1(AO3.4) [1]	FT from (b)

Question			Answer/Indicative content	Marks	Guidance
		d	Let the air resistance be $R \uparrow$ N2L $\uparrow$ gives $R + 140 - 15g = 15 \times (-1.5)$ $R - 7 = -22.5$ $R = -15.5$ Hence magnitude is 15.5 N R is downwards so motion of the package is upwards Alternative method Let the total upward force be F N and the air resistance $R \uparrow$ N2L $\uparrow$ gives $F - 147 = 15 \times (-1.5)$ So $F = 124.5$ Also $F = R + 140$ so $R = 124.5 - 140 = -15.5$ Hence magnitude is 15.5 N R is downwards so motion of the package is upwards	M1(AO3.3) M1(AO1.1) A1(AO1.1) A1(AO3.4) E1(AO3.4) [5] M1(AO3.3) A1(AO1.1) A1(AO1.1) A1(AO3.4) A1(AO3.4) [5]	Finding R using a and then T Finding F and hence R using a and then T
			Total	9	

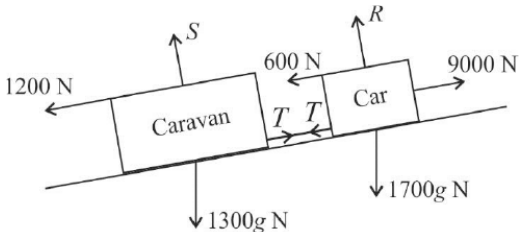
Question			Answer/Indicative content	Marks	Guidance	
13		a	E.g. The particle is in equilibrium [and the given forces cannot sum to zero as at 90°]	B1(AO2.2a) [1]	oe	Accept “without another force present, the particle would be moving on a rough surface without a frictional force”
		b	Friction 10 N [to give horizontal resultant of 0] Normal reaction from table. 8 N [to give vertical resultant of 0] Alternative method One extra force that gives equilibrium. Components 10 N  and 8 N  Components from Friction  and normal reaction 	B1(AO3.3) B1(AO1.2) B1(AO3.3) B1(AO1.2) [2]	oe Accept 'Because the surface is rough' for 'Friction' Oe oe Accept $\sqrt{164}$ at $\approx 39^\circ$ to horizontal oe Accept 'because the surface is rough' for 'Friction'	
			Total	3		

Question			Answer/Indicative content	Marks	Guidance
14		a	A Let acceleration be a in the direction of motion. N2L in direction of motion gives $-F = ma$ so $a = -\frac{F}{m}$, which is constant.	M1(AO3.3) A1(AO1.1)	Decide to use N2L to find acceleration No need to say 'constant'
		a	B (As a constant) use <i>suvat</i>, giving $S = 0 \times T - \frac{1}{2} \times \left(-\frac{F}{m}\right) T^2$ so $S = \left(\frac{F}{2m}\right) T^2$, and $k = \frac{F}{2m}$	B1(AO2.1) E1(AO2.4) [4]	Use appropriate (sequence of) <i>Suvat</i>
		b	As sliding, friction is limiting and $F = \mu R$ $R = mg$ $k = \frac{F}{2m}$ so $k = \frac{\mu mg}{2m}$ Hence $\mu = \frac{2k}{g} = \frac{2 \times 1.4}{9.8} = \frac{2}{7}$	M1(AO3.3) A1(AO3.4) M1(AO1.1) A1(AO2.2a) [4]	In $F = \mu R$, substitute for F & R in terms of m and g Or 0.286 (3s.f.)
Total				8	

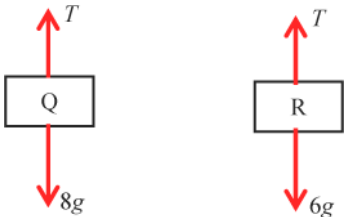
Question			Answer/Indicative content	Marks	Guidance	
15		a	Using N2L in direction of motion, acceleration $a \text{ ms}^{-1}$ $(40 - 10 - 5) = 10a$ so $a = 2.5$ Mark force in bar as tension $T \text{ [N]}$ Either For A $T - 10 = 6a$ or For B $40 - 5 - T = 4a$ so tension is 25 [N]	M1(AO3.3) A1(AO1.1) M1(AO3.4) M1(AO3.4) A1(AO1.1) [4]	$F = ma$ and all forces present Not required, may be implied Allow their a Allow their a Or $T = 25$	
		b	Without the force from the string, N2L in direction of motion $-15 = 10a^*$ so $a^* = -1.5$ Mark force in bar as tension $T^* \text{ [N]}$ either For A $T^* - 10 = 6a^* = -9$ or For B $-5 - T^* = 4a^* = -6$ so $T^* = 1$ giving a force of 1 [N] which is a tension	M1(AO3.3) A1(AO1.1) M1(AO3.4) M1(AO3.4) A1(AO1.1) A1(AO2.2a) [5]	Not required, may be implied Allow their a Allow their a Must be made clear in some way	
			Total	9		

Question			Answer/Indicative content	Marks	Guidance
16			$90 + 90 - 170 = 20a$ $a = 0.5$ $3 = 0 + \frac{1}{2} \times 0.5t^2$ $t = \sqrt{12}$ $v = 0 + 0.5\sqrt{12}$ $= \sqrt{3}$ In second phase resultant force is zero so $a = 0$ Time at constant speed $= \frac{7}{\sqrt{3}}$ $\text{Total time} = 2\sqrt{3} + \frac{7\sqrt{3}}{3} = 3.464... + 4.041... = 7.51 \text{ s}$ (3sf)	M1(AO 3.3) A1(AO 1.1b) M1(AO 1.1a) A1(AO 1.1b) M1(AO 3.4) A1(AO 1.1b) B1(AO 3.3) M1(AO 1.1a) A1(AO 1.1b) [9]	For use of Newton II For use of <i>suvat</i> with $s = 3$ and $u = 0$ 3.464... For use of <i>suvat</i> with their a and t to find v 1.732... Allow for 'constant speed' seen
			Total	9	

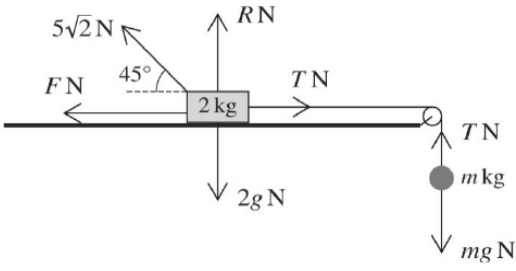
Question			Answer/Indicative content	Marks	Guidance	
17		a	$21 = \frac{1}{2}(u+0)14$ $u = 3 \text{ AG}$	M1(AO 1.1a) A1(AO 2.1) [2]	Use of <i>suvat</i> equation(s) to find u	
		b	$0 = 3^2 + 2a \times 21$ $a = -\frac{3}{14}$ Magnitude of force = $ ma = 18 \times \frac{3}{14}$ $= \frac{27}{7} = 3.86 \text{ N}$	M1(AO 3.1b) M1(AO 1.1a) A1(AO 1.1b) [3]	Use of <i>suvat</i> to find acceleration Use of Newton II Must be magnitude (positive)	
		c	$N = 18g$ $3.86 = \mu \times 18g$ $\mu = 0.022 \text{ to } 2 \text{ sf}$	B1(AO 3.1b) M1(AO 3.4) A1(AO 1.1b) [3]	soi Use of $F = \mu N$ to find μ Must be given correct to 2sf	
			Total	8		

Question			Answer/Indicative content	Marks	Guidance
18		a	 Weights and normal reactions for each correctly shown 9000 N and force T in coupling correctly shown Resistances 600 N and 1200 N correctly shown	B1(AO 1.1a) B1(AO 3.3) B1(AO 3.3) [3]	The two force diagrams may be shown separately
		b	Resolve weights parallel to the slope $9000 - 600 - 1200 = 3000g \sin 8 - 3000a$ $7200 - 4091.689 = 3000a \Rightarrow a = 1.04 \text{ m s}^{-2}$	M1(AO 3.1a) M1(AO 3.3) A1(AO 1.1b) A1(AO 1.1b) [4]	Allow sin/cos errors here Newton II; all non-weight terms required All terms correct cao

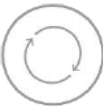
Question			Answer/Indicative content	Marks	Guidance
		c	$T - 1200 - 1300g \sin 8 = 1300 \times 1.04$ or $9000 - 1700g \sin 8 - 600 - T = 1700 \times 1.04$ $T = 4320 \text{ N}$ (answer using a unrounded)	M1(AO 3.4) A1(1.1a) A1FT(AO 1.1b) [3]	Resolving parallel to the slope for car or caravan; all forces present, correct mass All terms in equation correct (for their a) FT their rounded / unrounded a
			Total	10	

Question			Answer/Indicative content	Marks	Guidance
19		i		B1 [1]	The same symbol for T must be used in both diagrams. Examiner's Comments Part (i) asked for force diagrams for the two connected particles, in this case blocks. A minority of candidates failed to get this right; the most common error was to mark different tensions for the different parts of the string.
		ii	Q: $8g - T = 8a$ R: $T - 6g = 6a$	B1 B1 [2]	Allow the equivalent equations with the direction of a reversed Examiner's Comments Part (ii) followed on from part (i) with a request for the equations of motion of the two blocks. This was not universally well answered; sign errors were common.

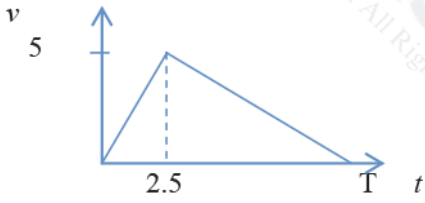
Question			Answer/Indicative content	Marks	Guidance
		iii	Adding the equations of motion $2g = 14a$ $a = \frac{2g}{14} \quad (= 1.4 \text{ m s}^{-2})$ For Q: $s = ut + \frac{1}{2}at^2$ $2 = \frac{1}{2} \times 1.4 \times t^2$ $\Rightarrow t = 1.690\dots$ so the time is 1.69 s	M1 A1 M1 A1 [4]	Eliminating one variable from the two equations. May be implied by subsequent working. This answer must be consistent with the direction of a used in part (ii) Or an equivalent sequence of constant acceleration formulae Dependent on previous M mark. FT for their a but do not allow if it is g CAO Examiner's Comments In part (iii) the blocks were released and candidates were asked to find the time taken for one of them to reach the floor. This involved finding the acceleration of the system. Many, including those who has made mistakes in the earlier parts, did this using a whole system approach rather than working from the equations of motion.
			Total	7	
20			Resolve down the slope, using Newtons' second law $mg \sin 20^\circ = ma$ $v^2 = 0^2 + 2(g \sin 20^\circ)s$ $v = \sqrt{13.407\dots} = 3.66 \text{ m s}^{-1}$	M1(AO3.3) A1(AO1.1b) M1(AO1.1a) A1(AO1.1b) [4]	Use of $v^2 = u^2 + 2as$
			Total	4	

Question			Answer/Indicative content	Marks	Guidance	
21		a		B1(AO1.1a) B1(AO3.3) [2]	Weights and normal reaction labelled Tension correct in both parts of the string and friction in the correct direction	Condone absence of units N in diagram Friction force could be shown as μR or $0.6R$ and tension could be shown as mg
		b	$R + 5\sqrt{2} \sin 45^\circ = 2g$ $R = 2g - 5 (= 14.6)$ $F = 0.6(2g - 5)$ $F = 8.76 \text{ N}$	M1*(AO3.4) A1(AO1.1b) M1(AO1.1a)dep* A1(AO1.1b) [4]	Resolve vertically for the block Numerical evaluation not needed here Use of $F = \mu R$, but do not allow if $R = 2g$	Allow sin or cos here
		c	$T = F + 5\sqrt{2} \cos 45^\circ (= 13.76)$ $T = mg$ $m = \frac{13.76}{g} = 1.40$	M1(AO3.4) A1(AO1.1b) M1(AO1.1a) A1(AO1.1b) [4]	Resolve horizontally for the block Correct equation; evaluation not required for this mark Resolve vertically for the ball	Allow sin or cos here soi
			Total	10		

Question			Answer/Indicative content	Marks	Guidance
22		a	$v^2 = u^2 + 2as$ $1.2^2 = 2 \times a \times 1.8$ $a = 0.4 \text{ ms}^{-2}$	M1 (AO 3.3) A1 (AO 1.1b) [2]	Using suitable suvat equation(s) leading to value for a <u>Examiner's Comments</u> Most candidates successfully used the suvat equations to reach the correct answer.
		b	$F - 19 = 2.8a$ $F = 20.12$ (20.1 N to 3sf)	M1 (AO 3.3) A1 (AO 1.1b) [2]	Using Newton's second law. All terms present. Allow 20 N FT their a <u>Examiner's Comments</u> Newton's second law was well understood and most candidates were successful here.
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
23			$T - \frac{1}{2}mg = 4ma$ $mg - T = ma$ $a = \frac{g}{10} \text{ m s}^{-2}, T = \frac{9mg}{10} \text{ N}$	B1 (AO3.1b) B1 (AO3.1b) M1dep (AO1.1a) A1 (AO1.1b) [4]	Award for first of these two equations Signs consistent with first equation Attempting to solve the simultaneous equations leading to either T or a so Both T and a needed simplified M1 dependent on at least one B1 earned Allow 0.98 m s^{-2}, and $T = \frac{441m}{50} = 8.82m \text{ N}$ Examiner's Comments This question was a very standard question but many candidates did not correctly use Newton's 2nd law to form equations of motion. Those who attempted to write down a single equation for the whole system (the round-the-corner method) were rarely successful. Some wrongly included the weight of the object on the table in its equation of motion. Many candidates had fragments of working that were not clear - examiners used the mass to indicate which part(s) of the system was being considered and required the correct forces acting on that part. Some who had correct equations then lost a mark as they did not simplify their expressions for a and T fully.  Prepare candidates to consider each part of the system separately and to identify which forces are acting on that object in the direction of its motion.  There is evidence of candidates confusing mass and weight, essentially using $F = mga$ instead of Newton's 2nd law.
			Total	4	

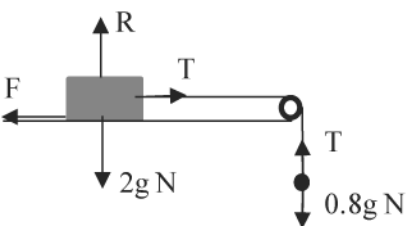
Question			Answer/Indicative content	Marks	Guidance	
24		a	Use of Newton's 2nd Law for the whole train Weight component parallel to the plane is $0.5g \sin 20^\circ$ $2.5 - 0.5g \sin 20^\circ = 0.5a$ $a = 1.65 \text{ ms}^{-2}$ to 3sf Alternative solution Equations for engine and truck separately N2L for engine: $2.5 - 0.3g \sin 20^\circ - T = 0.3a$ N2L for truck: $T - 0.2g \sin 20^\circ = 0.2a$ $a = 1.65 \text{ ms}^{-2}$ to 3sf	B1 (AO 3.3) M1 (AO 1.1a) A1 (AO 2.1) B1 M1 A1 [3]	May be implied by use in an equation N2L with at least one correct force AG; must be clearly shown Correct weight component in at least one equation Both equations attempted, plus attempt to eliminate T AG; must be clearly shown	$m = 0.5$ used $m = 0.3$ and/or 0.2 used
		b	Either $T - 0.2g \sin^\circ = 0.2 \times 1.65$ Or $2.5 - 0.3g \sin 20^\circ - T = 0.3 \times 1.65$ $T = 1 \text{ N}$	M1 (AO 3.3) A1 (AO 1.1b) [2]	Use of N2L for either engine or truck Allow awrt 1.00	Equation must have all the forces and no extras
		c	$3^2 = 1^2 + 2 \times 1.65 \times s$ $s = 2.42 \text{ m}$	M1 (AO 1.1a) A1 (AO 1.1b) [2]	Use of suvat leading to a value for s Or 2.43 from use of $a = 1.6482...$	
			Total	7		

Question	Answer/Indicative content	Marks	Guidance
25	EITHER acceleration phase $v^2 = 0 + 2.5 \times 2 = 5 \text{ m s}^{-1}$ slowing phase $v^2 = u^2 + 2as$ $0 = 5^2 + 2a \times 10$ $a = -1.25 \text{ m s}^{-2}$ $[-R] = 1.5 \times (-1.25) = -1.875$ Magnitude of $R = 1.875 \text{ N}$ (1.88 to 3sf) OR acceleration phase $v = 0 + 2.5 \times 2 = 5 \text{ m s}^{-1}$  using the distance to find the time it takes to stop <i>using areas</i> (second triangle): $\frac{1}{2}(T - 2.5) \times 5 = 10$ $T = 6.5 \text{ so time to stop is } 4 \text{ s.}$ So $0 = 5 + 4a$	M1 (AO3.1b) A1 (AO1.1b) M1 (AO3.1b) A1 (AO1.1b) M1 (AO1.1a) A1 (AO1.1b) [6] M1 A1 M1 A1	Use of <i>suvat</i> equation(s) to find velocity. Do not allow if $s = 10$ used Use of <i>suvat</i> equation(s) with $s = 10$ to find acceleration FT their velocity. Must be correct sign. Use of Newton's second law. FT their a $a \neq 2$ Must be positive Use of <i>suvat</i> equation(s) to find velocity. Do not allow if $s = 10$ used Use of area and <i>suvat</i> equation(s) to find acceleration Must be correct sign Must recognise two phases of motion for first 4 marks Consistent sign convention needed for full credit. Must recognise two phases of motion for first 4 marks

Question			Answer/Indicative content	Marks	Guidance	
			Giving $a = -1.25 \text{ m s}^{-2}$ $[-R] = 1.5 \times (-1.25) = -1.875$ Magnitude of $R = 1.875 \text{ N}$ (1.88 to 3sf)	M1 A1 [6]	Use of Newton's second law. FT their $a \neq 2$ Must be positive <u>Examiner's Comments</u> Many good clear solutions were seen, however some candidates did not realise that this question covered two phases of motion and used all the numbers in the question in a single set of suvat equations. Some simply extracted the values of mass and acceleration from the first phase of motion and multiplied them together. Some candidates who obtained a negative value for resistance did not notice that it was the magnitude of the resistance that was required, so a positive answer was needed.  Look out for two phases of motion and set up different equations for the two phases. Use the value of the velocity at the end of the first phase to link the two phases.	Consistent sign convention needed for full credit.
			Total	6		

Question		Answer/Indicative content	Marks	Guidance	
	b	N2L for truck A: $T_1 - T_2 - 600 = 25\,000 \times 0.1$ Difference $T_1 - T_2$ is 3100 N Alternative solution For the engine $D - 2000 - T_1 = 8000$ $T = 6200$ N For truck B $T_2 - 600 = 2500$ $T_2 = 3100$ N Difference $T_1 - T_2$ is 3100 N	M1 (AO3.1b) A1 (AO1.1) [2] M1 A1 [2]	All forces present and no extras Signs must be consistent from correct working All forces present and no extras in one equation leading to a value for either T_1 or T_2. Signs must be consistent FT their D	Values for tensions need not be found The equations may have been found in part (b) Also allow T_1 is twice T_2 or from correct values
		Total	4	Examiner's Comments Many candidates were able to achieve the method mark here for considering truck B even where their driving force had been incorrect. The expected method was to find the difference between the two tensions. Full credit was allowed for saying that one tension was twice the size of the other as the question asked candidates to "determine how much larger" rather than "find the difference".	

Question			Answer/Indicative content	Marks	Guidance	
27			N2L gives $4000 - 800 = 1200a$ $a = \frac{8}{3} \text{ m s}^{-2}$ Using $v = u + at$ gives $v = \frac{8}{3} \times 9 = 24 \text{ m s}^{-1}$	M1 (AO 3.1b) A1 (AO 1.1b) M1 (AO 3.3) A1 (AO 1.1b) [4]	Both forces present; allow sign errors Soi Using <i>suvat</i> equation(s) leading to a value for v FT their a Examiner's Comments A few candidates were not clear about the distinction between mass and weight and used $F = mga$ for Newton's second law. The majority of candidates knew how to calculate the acceleration and use it to find the velocity. Some rounded or truncated their acceleration, but rarely so much that their final answer did not agree with the given answer to at least 2 significant figures.	Allow for $3200 = 1200a$ seen
			Total	4		

Question			Answer/Indicative content	Marks	Guidance	
28	a			B1(1.1a) B1(1.1a) B1(1.1a) [3]	both weights correct. common tension in the right directions Friction and normal reaction and no extra forces	Allow weight of box and weight of sphere, but not if both marked weight. Allow T_1 and T_2 provided they are clearly shown equal to each other elsewhere. For F, allow $0.35R$, $0.35 \times 2g$ $0.7g$ or 6.86 N
	b		$T - F = 2a$ $0.8g - T = 0.8a$	B1(AO1.1a) B1(1.1a) [2]	Allow any expression for F Allow distinct tensions if consistent with diagram	For F, allow $0.35R$, $0.35 \times 2g$ $0.7g$ or 6.86 N
	c		Vertically for the block $R = 2g$ Friction $F = 0.35R = 0.7g$ Add equations $0.8g - F = 2.8a$ $0.8g - 0.7g = 2.8a$ $a = 0.35 \text{ m s}^{-2}$	M1(AO3.1b) A1(AO2.1) M1(AO2.1) A1(2.1) [4]	Attempt to use μ to evaluate friction Correct value for F Eliminate T from their equations AG must follow from correct work	Some of this work may already have been seen in previous part
	d		Use $s = 0.5$, $u = 0$, $a = 0.35$ $0.5 = \frac{1}{2} \times 0.35 \times t^2$ $t = 1.69 \text{ s}$	M1(AO1.1a) A1(AO1.1) [2]	Using $suvat$ equation(s) leading to a value for t Do not allow ± 1.69	
			Total	11		

Question			Answer/Indicative content	Marks	Guidance	
29			$u_x = 35 \cos \theta$ giving $x = (35 \cos \theta)t$ $u_y = 35 \sin \theta$ giving $y = (35 \sin \theta)t - \frac{1}{2}gt^2$ Substitute for t $y = (35 \sin \theta) \left(\frac{x}{35 \cos \theta} \right) - \frac{1}{2}g \left(\frac{x}{35 \cos \theta} \right)^2$ $\left[y = x \tan \theta - \frac{x^2}{250 \cos^2 \theta} \right]$	B1(AO3.3) B1(AO3.3) M1(AO3.3) A1(AO1.1) [4]	Award seen in any form soi Substituting for t in their equation for y Award in any form ISW	
			Total	4		
30			Newton's second law $\begin{pmatrix} 0 \\ -1.5g \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix} + \begin{pmatrix} 2 \\ 20 \end{pmatrix} = m\mathbf{a}$ $\mathbf{a} = \frac{1}{1.5} \begin{pmatrix} 6 \\ 3.3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2.2 \end{pmatrix} \text{ m s}^{-2}$	M1(AO1.1a) A1(AO1.1) [2]	Addition of vectors. Allow if weight missing but other two forces and acceleration seen in equation Must be vector; any correct form.	Allow wrong weight only if given as a vector
			Total	1		